

01

Proofs

Combine quantifier rules with the proof steps we already know. Find the right proof type for your problem.



EXERCISE 1

Write Complete Proof using Rules of Inference.

I eat spinach (S) or ice cream (I).

If I study logic (L), then you pass the exam (E).

If I eat ice cream, then I study logic.

If I eat spinach, then I play golf (G).

You fail the exam.

01 therefore I play golf.

Yes. Translate first: $S \vee I \quad L \rightarrow E \quad I \rightarrow L \quad S \rightarrow G \quad \neg E$.

$\neg E \quad L \quad I$ (modus tollens, twice).

I with $S \vee I$ gives S (disjunctive syllogism).

$S \rightarrow G$ (modus ponens).



RETRIEVAL

So far

Lecture 6: valid arguments, named rules, and justified proofs.

p

Valid arguments

$q \quad p$

No row makes every premise true
and the conclusion false.

q

**MP, MT,
HS, DS**

Implication & OR rules

Modus ponens, modus tollens,
syllogisms, resolution.

p

AND rules

$q \quad p$

Simplification and conjunction
split and combine facts.

Lecture 7 assumes fluency in each of these rules.

$\therefore$

Building & auditing proofs

Justify every line; catch the
consequent and antecedent
errors.

THE ROUTE

Today

Extend inference to quantifiers,
then choose a proof strategy.

01

APPLY

Quantified rules

Instantiate, generalize, and introduce a fresh witness.

02

EVALUATE

Contraposition

A parity proof shows when the reversed route is easier.

P Q
 Q P

03

CREATE

Direct proof & cases

Build from definitions, or exhaust every possibility.

p q / C
 C

04

CREATE

Contradiction

Assume the denial and derive an absurdity: $\sqrt{\quad}$,
then primes.

QUANTIFIED INFERENCE

Universal instantiation

Universal = for all. Instantiation = taking a particular instance.

$$x P x$$

c belongs to the domain

$$\therefore P c$$

KNOW

Every lab robot is registered.

CHOOSE

Alice is a lab robot.

CONCLUDE

Alice is registered.

A universal claim applies to each allowed object.

INstantiate the whole formula

The registered robot

LINE	STATEMENT			REASON
1	x	x	x	Premise
2				Premise
3				1: universal instantiation
4				2, 3: modus ponens

The domain is all objects. identifies the robots.

QUANTIFIED INFERENCE

Universal generalization

Universal = for all. Generalization = forming a general claim.

$P\ c$ proved

c arbitrary

$\therefore x\ P\ x$

ARBITRARY INTEGER

Let c be any integer.

PROVE

$$c + 0 = c.$$

GENERALIZE

Every integer x satisfies $x + 0 = x$.

No undischarged assumption may single out c .

QUANTIFIED INFERENCE

Existential generalization

Existential = there exists. Generalization = forming a general claim.

$$P\ c$$

c belongs to the domain

$$\therefore \exists x\ P\ x$$

A WITNESS

Alice solved the maze.

CONCLUSION

Someone solved the maze.

MEANING

At least one is enough.

A known instance provides an existential witness.

EXERCISE 2

S : submitted. T : tested. E : eligible.

Domain: students. $x \quad S \quad x \quad T \quad x \quad E \quad x \quad S \quad T$.

01 Apply Universal Instantiation:

$S \quad T \quad E$.

02 Apply Conjunction:

$S \quad T$, by conjunction.

03 Apply Modus Ponens:

E , by modus ponens.

QUANTIFIED INFERENCE

Existential instantiation

Existential = there exists. Instantiation = introducing a particular instance.

$$x P x$$

Choose fresh c : $P c$

EXISTENCE

Some robot solved the maze.

A TEMPORARY NAME

Call a qualifying robot c .

RESTRICTION

c is not a preselected robot.

Keep witness-dependent reasoning within its scope.

EXERCISE 3

True or False: one line justification!

Domain: people. $P x$: plays piano. $Q x$: codes.

01 $x P x \quad x Q x \quad \therefore x P x \quad Q x ?$

Not guaranteed. Alice only plays; Bob only codes.

02 $x P x \quad Q x \quad \therefore x P x ?$

Valid. Choose a fresh joint witness, simplify, then generalize.

LINE	STATEMENT	REASON
1	$x R x \quad C x$	Premise
2	$x C x \quad A x$	Premise
3	$x R x$	Premise
4	$R c$	3: existential instantiation
5	$R c \quad C c$	1: universal instantiation
6	$C c$	4, 5: modus ponens
7	$C c \quad A c$	2: universal instantiation
8	$A c$	6, 7: modus ponens
9	$R c \quad A c$	5, 7: hypothetical syllogism
10	$x A x$	8: universal generalization

Given the first three premises, audit every remaining line.

Line 10 is illegal; lines 1-9 are all correct.

02

Proof by contraposition

The same conditional can offer a different route.



P Q
 Q P

DEFINITIONS FOR THE NEXT PROOF

Even and odd integers

Even

$$n = 2k$$

For some $k \in \mathbb{Z}$. An integer multiple of two.

Odd

$$n = 2k + 1$$

For some $k \in \mathbb{Z}$. One more than an even integer.

Every integer is exactly one of even or odd.

EXERCISE 4

A parity claim

For $n \in \mathbb{Z}$, prove: if n is odd, then n is odd.

01 What is the contrapositive?

If n is even, then n is even.

02 Which definition starts that proof?

Write $n = k$ for an integer k .

CONTRAPOSITION

The parity proof

STEP	STATEMENT	REASON
1	Let $n \in \mathbb{Z}$ be arbitrary.	Set the domain
2	Assume n is even: $n = 2k$, $k \in \mathbb{Z}$.	Negate the original conclusion
3	$n = 2k$	Substitution
4	n is even	Algebra
5	$n = 2k$, so n is even.	Definition of even
6	Thus n odd implies n odd.	Contraposition

Only n even was assumed for the contrapositive.

EXERCISE 5

A new parity proof

For $n \in \mathbb{Z}$, prove: if n is odd, then n is odd.

01 State the contrapositive.

If n is even, then n is even.

02 Complete the proof.

$n = 2k$ gives $n = 2k = 2k$.

PROOF METHODS

What counts as a proof?

A valid argument establishing a mathematical statement.

Axiom / postulate

Accepted foundation.

Hypothesis / premise

Assumed for this argument.

Theorem

Established by a proof.



A conjecture is a claim still awaiting proof.

PROOF METHODS

Divides means “is a multiple of”

Here $m, n \in \mathbb{Z}$ and $m \neq 0$.

$m \mid n$

$n = km$ for some $k \in \mathbb{Z}$



because $n = km$.

WORKED EXAMPLE • TRANSITIVITY

A multiple of a multiple

Given $a \mid b$ and $b \mid c$, prove $a \mid c$.

Unpack

$$b = ra \quad c = sb$$

$$r, s \in \mathbb{Z}$$

Substitute

$$c = s(ra) = (sr)a$$

Finish

$$sr \in \mathbb{Z} \quad a \mid c$$



EXERCISE 6

Keep the common divisor

Let $d \mid a$ and $a \mid b$. For any $x, y \in \mathbb{Z}$, prove:

$$d \mid xa + yb$$

Write $a = rd$ and $b = sa$, with $r, s \in \mathbb{Z}$.

$$xa + yb = xrd + ysr = d(xr + ys).$$

The coefficient $xr + ys$ is an integer.



PROOF METHODS

Prime or composite?

For an integer n :

Prime

Positive divisors: 1 and n

Example: 7

Composite

A divisor m with $m \neq n$

Example:

1 is neither prime nor composite.

WORKED EXAMPLE

One factor settles 289

So 289 is composite.



One proper divisor is enough.